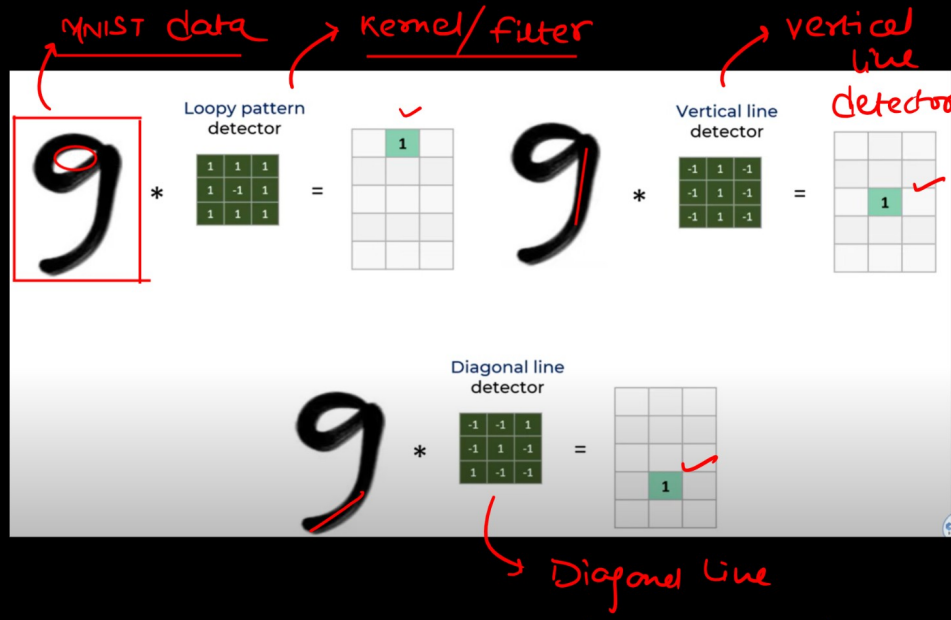
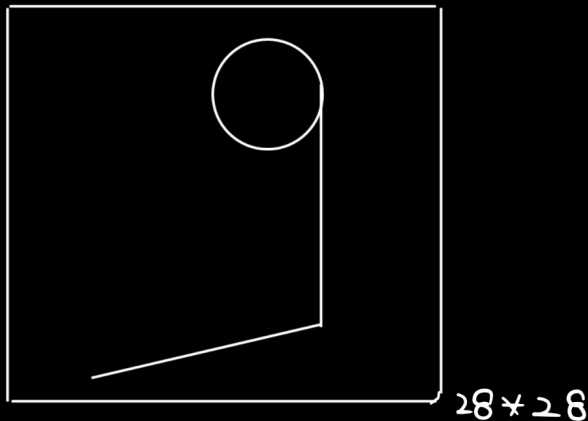
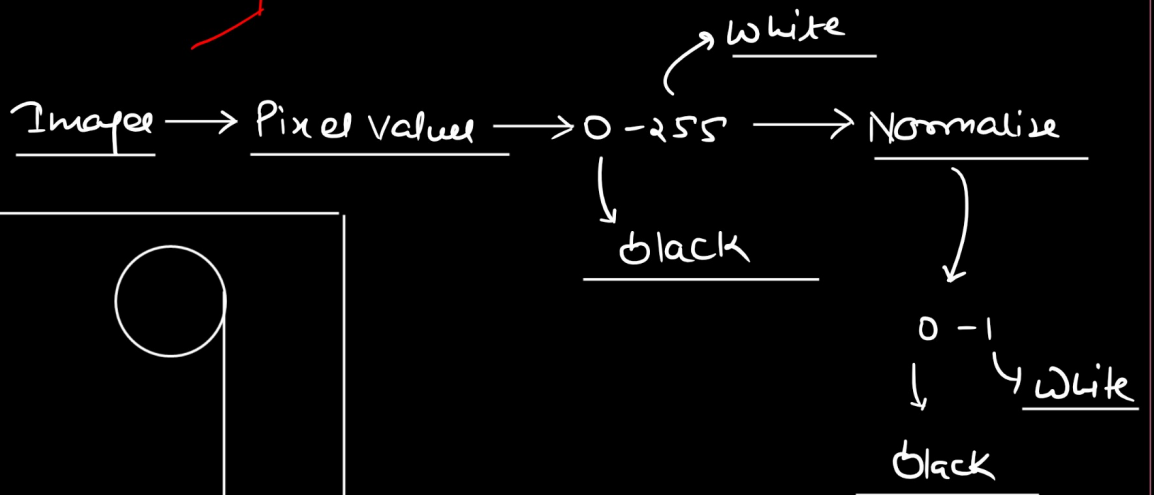
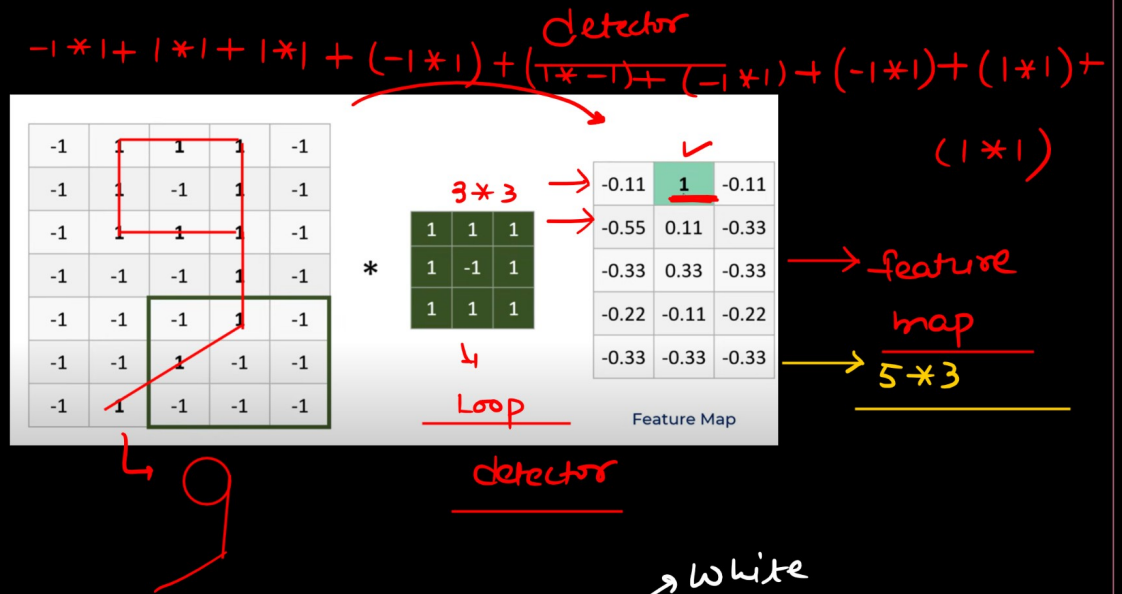


CNN (Convolutional Neural Network)

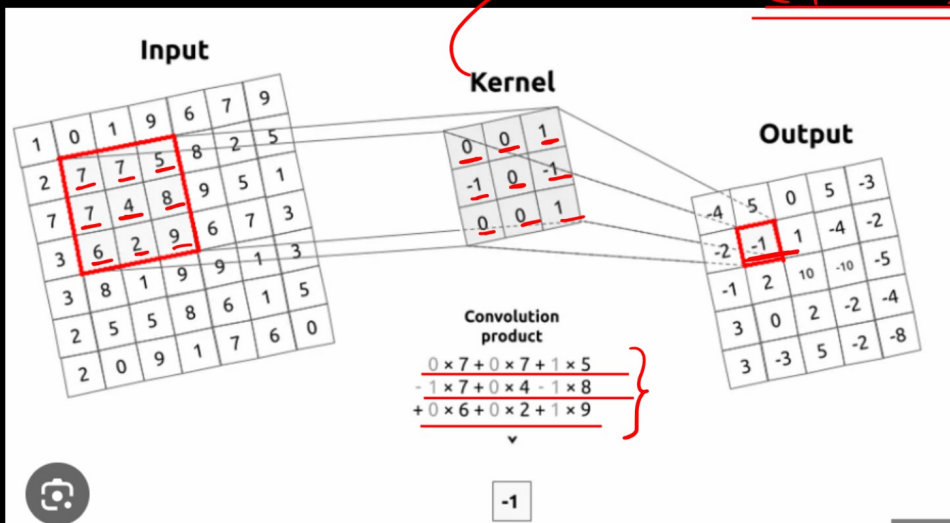


- ① Loopy pattern
- ② Vertical line
- ③ Diagonal line

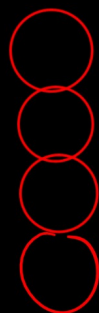
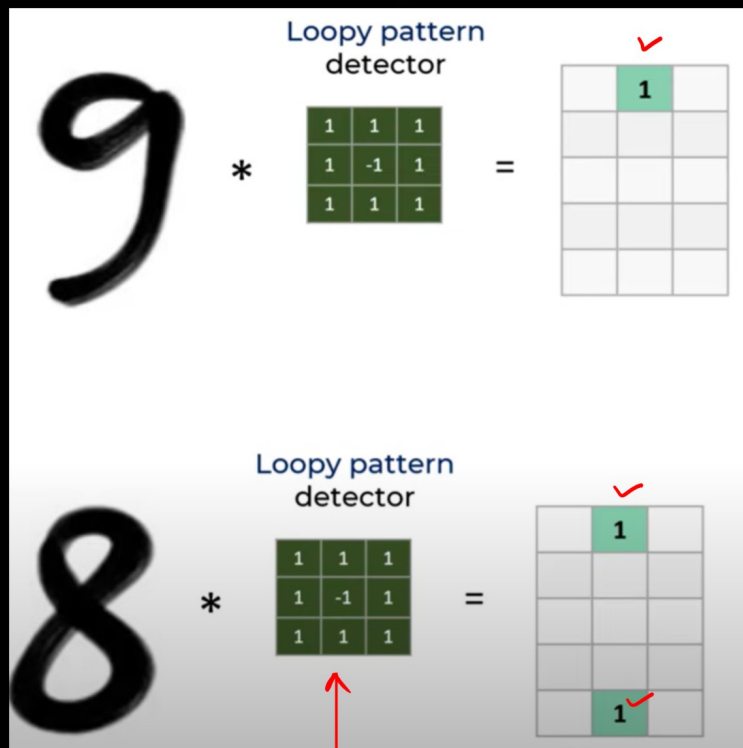
7*5



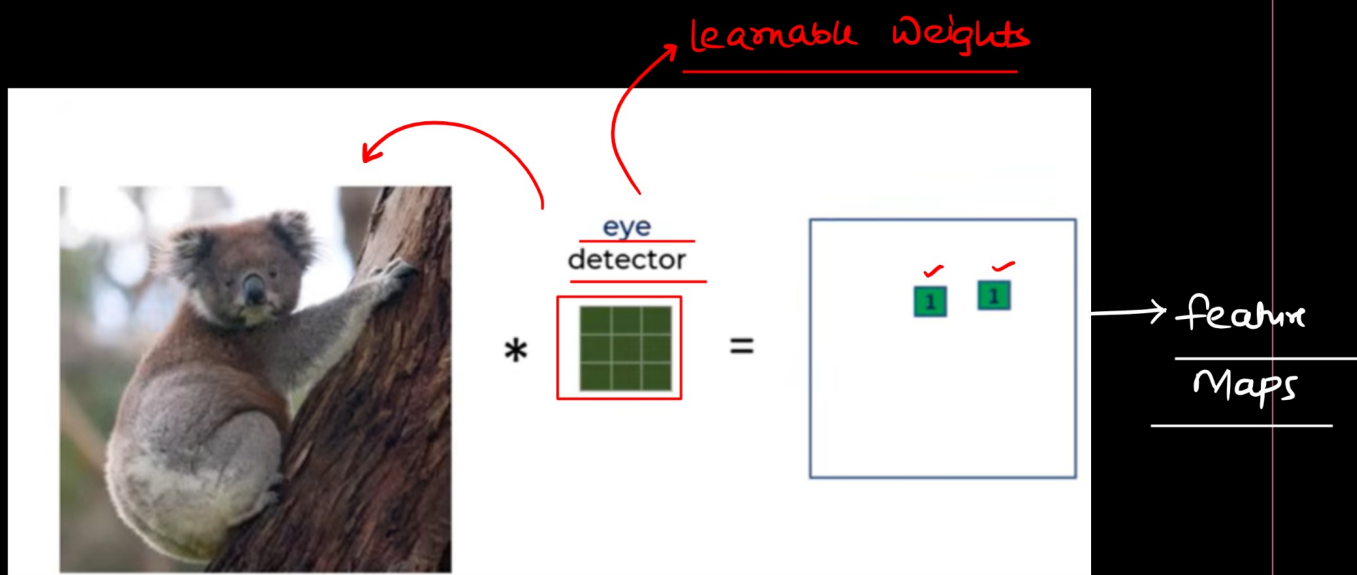
Predefined (operation)



Important patterns in a given image



	1	
	1	
	1	
	1	



$$\frac{N - f + 2P}{s} + 1$$
 where:

- N = input image size
- f = filter size
- P = padding = 0
- s = stride

 → num of jumps = 1

$$\frac{7 - 3 + 2 \times 0}{1} + 1$$

$$\Rightarrow 5 \text{ (num of rows)}$$

Kernel $\rightarrow 3 \times 3$
 I/p shape $\rightarrow 7 \times 5$
 stride $\rightarrow 1$

padding $\rightarrow 0$

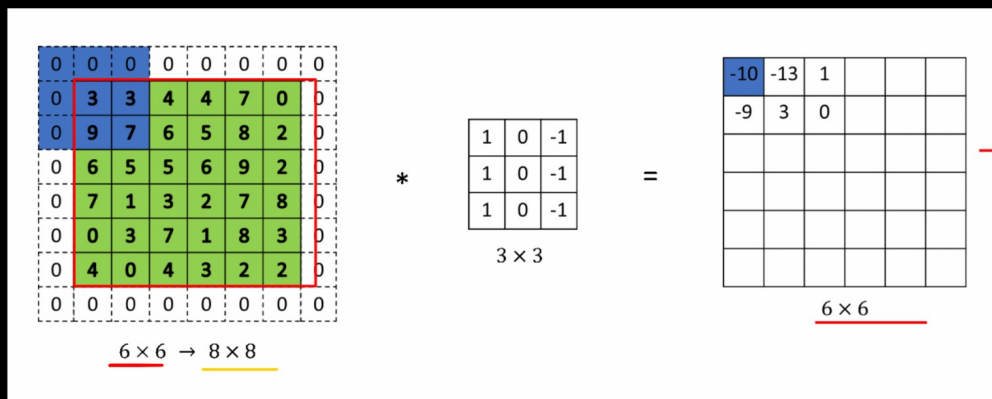
$$\frac{5 - 3 + 2 \times 0}{1} + 1 \Rightarrow 3 \text{ (num of cols)}$$

Output shape $\Rightarrow 5 \times 3$

Input image shape = Output image shape

$P=1$

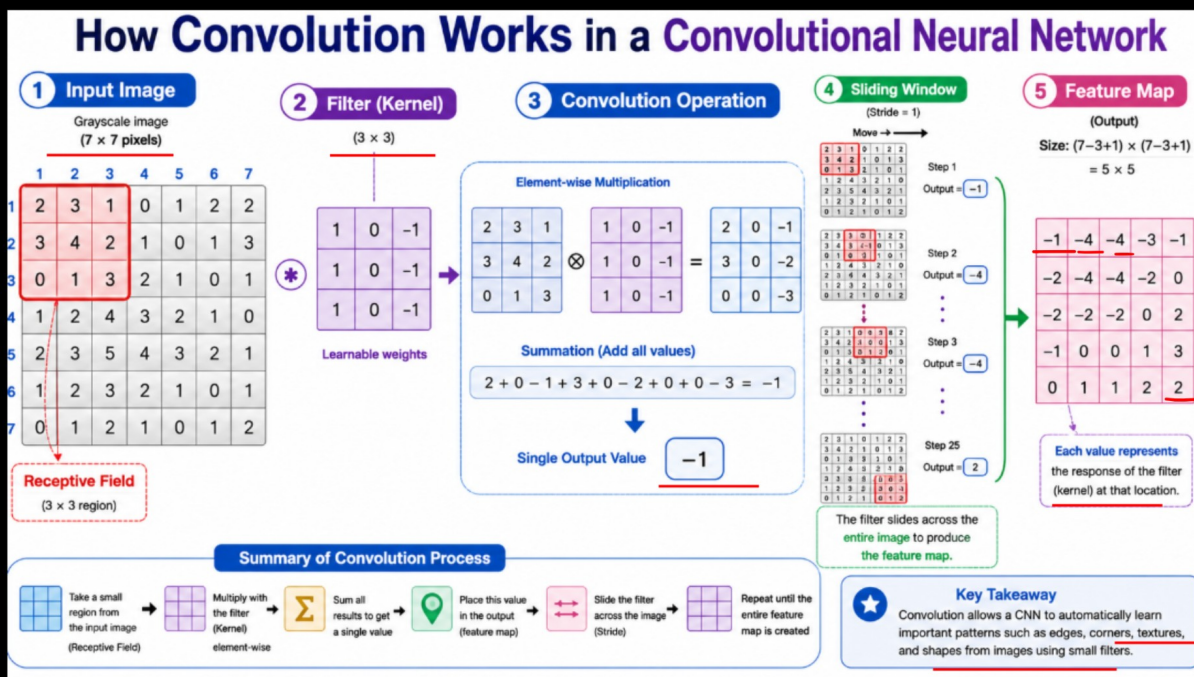
Requirement



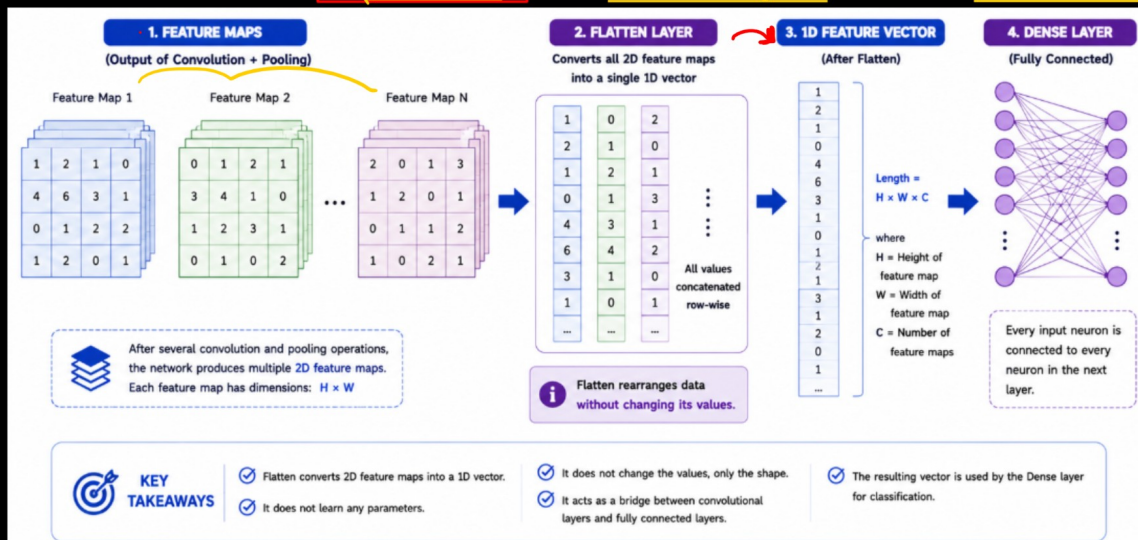
Input image = 6x6

$$6 - 3 + 2 \times 1 + 1 \Rightarrow 6$$

output image = 6x6

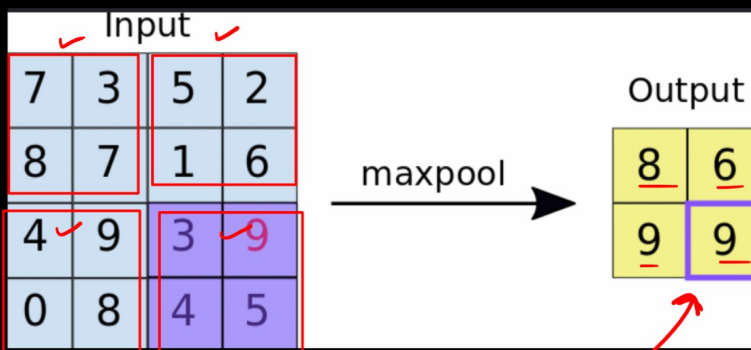


CNN → Input layer → Hidden layer



Output layer

Maxpooling → take all the necessary/higher pixel values



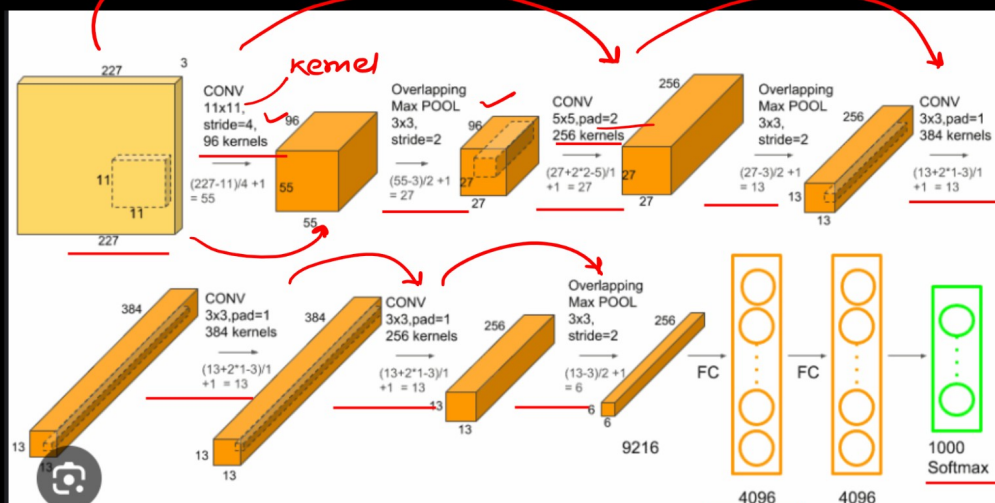
Downsampling

(less computation resource/faster training)

AlexNet

$$\frac{227 - 11 + 2 \times 0}{4} + 1$$

$$227 \times 227 \times 3 \Rightarrow 55$$



built the alexnet architecture from scratch

ImageNet → { VGGNet, ResNet, InceptionNet,
MobileNet & many more }